

FIFTH SEMESTER DIPLOMA EXAMINATION IN AUTOMOBILE ENGINEERING

(MODEL QUESTION PAPER)

Automobile Transmission

(Maximum Mark 100)

Time 3 hours

PART A

I (Answer the following questions in one or two sentences)

1. Name any two clutch facing materials
2. Define gear ratio
3. State the use of slip joint in propeller shaft
4. Name 2 different types of final drives
5. 6.00x16 indicates a tyre size. What is 6.00 and 16? (5x2=10Marks)

PART B

II (Answer any 5 questions. Each question carries 6 marks)

1. Explain the construction and working of Diaphragm clutch.
2. Syncromesh gear box is more efficient than Constant mesh gear box. Justify the statement with 6 points.
3. A KSRTC bus driver seems to disengage the clutch twice for changing gears .Explain the situation as a technician. What is its necessity?
4. Sketch and explain the arrangements of Torque tube drive type propeller shaft.
5. Sketch and explain a semi floating type rear axle.
6. Write short note on 1) Tread 2) Bead 3) Shoulder
7. List 6 desirable properties of a tyre.

(5x6=30 Marks)

PART C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT -1

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| III. | (a) Explain the working of a fluid flywheel with a neat figure | 8 |
| | (b) Explain the working of Vacuum operated clutch. | 7 |

OR

- IV. (a) Explain 4 major clutch troubles. How these can be remedied? 8
(b) Write short note on 1)Release bearing 2) Free pedal play 7

UNIT -2

- V. (a) Describe the working of a 4 speed synchromesh gear box with a sketch. 8
(b) Discuss the interlocking method of gear shifting mechanism. 7

OR

- VI. (a) Explain the constructional features of a planetary gear set. 8
(b) Explain the working of an over drive mechanism. 7

UNIT-3

- VII. (a) Compare banjo and split type rear axle casing. 8
(b) Explain the arrangement of two- piece propeller shaft. 7

OR

- VIII. (a) Describe the working of any two variable velocity joints. 8
(b) Explain the method of operation of differential lock with a sketch 7

UNIT-4

- IX. (a) Write brief description about different processes involved in tyre manufacturing 8
(b) Discuss various effects of under and over inflation of a tyre 7

OR

- X. (a) Describe the construction of disc wheel with a sketch 8
(b) write brief description about 1)Cross ply 2) Radial ply 7